MATTHEW POOL

Software Engineer | Product Designer

UI/UX | Native Mobile | Full-Stack Web | AI/ML | B2B/B2C

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🌐 matthew-pool.github.io

PROFESSIONAL SUMMARY

Summa Cum Laude Computer Science graduate (3.99 GPA) and Software Engineer with a rare background spanning end-to-end product development, five years as a top-ranked Fortune 500 B2B/B2C sales agent, and 10+ years in customer service and technical support. I design in Figma, architect the backend, ship to production, and understand exactly what makes users stay. I bridge the gap between UI/UX design and technical implementation throughout the entire product lifecycle — delivering accessible, high-ROI software that solves real user problems.

EDUCATION

Bachelor of Science in Computer Science

Southern New Hampshire University (Manchester, NH) • Sep 2021 – Dec 2024

🏆 **Summa Cum Laude** • ***3.99 GPA*** • **President's List (2023–2024)**

📜 ***Certificate in Data Analysis*** • *uCertify (2024)*

Associate of Arts in General Studies

Tyler Junior College (Tyler, TX) • Aug 2003 – May 2005

TECHNICAL SKILLS

🎨 Product Design

UI/UX Design, Figma, Inkscape, GIMP, Adobe Creative Suite, CAD, SketchUp, Responsive Design, User Flows, Information Architecture

💻 Programming Languages

Python, JavaScript, TypeScript, Kotlin, Java, C/C++, C#, SQL

📱 Mobile Development

Android SDK, Jetpack Compose, Material Design 3, MVVM Architecture, Coroutines, Hilt, Firebase, REST APIs

🌐 Web Technologies

React, Angular, Node.js, Express.js, LoopBack API, RxJS, HTML5, CSS3, Vite, MVC, Bootstrap, Tailwind CSS, Handlebars, Spring Boot

🗄️ Databases

MySQL, MongoDB, Mongoose (ODM), AWS DynamoDB, Firebase Firestore

🤖 Data Science & Machine Learning

TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, DNN, CNN, Deep Q-Networks (DQN)

🎮 Gaming & Graphics

Unity, OpenGL/GLSL, Reinforcement Learning (RL), Deep Q-Networks (DQN)

🧪 Testing & QA

Postman, pytest, JUnit, MockK, Mockito, Espresso, RoboElectric, Jacoco, Jest, Selenium

⚙️ DevOps & Security

Docker/Docker Compose, AWS, CI/CD, Git/GitHub Actions, JWT, OWASP, Snyk, TLS/SSL, RSA/AES, Gradle, Maven

TECHNICAL PROJECTS

Flick — Android Streaming App ↓ *full details in Experience*	*Kotlin, Jetpack Compose, MVVM, Hilt, Firebase, Archive.org API, GitHub Actions*	Proprietary ML recommendation engine (cosine similarity + TF-IDF), dual-axis gesture navigation, and a CI/CD pipeline via GitHub Actions. Currently in closed testing on Google Play.
Full-Stack Travel Platform	*Angular, TypeScript, LoopBack API, Docker, AWS DynamoDB*	Engineered an Angular/TypeScript SPA with a Docker-containerized backend, orchestrating a full database migration from MongoDB to AWS DynamoDB.
The Existence Paradox	*React, TypeScript, Tailwind CSS, Vite*	Interactive responsive web application featuring dynamic state management and a complex decision-tree navigation system with branching logic.
Financial Security Audit & Remediation	*Java, Spring Boot, OWASP, AES-256, SHA Cryptography*	Refactored a legacy RESTful API to implement AES-256 encryption at rest, mitigating injection vulnerabilities and achieving full OWASP compliance.
Austin Animal Center Dashboard	*Python, MongoDB, Dash, Pandas, Plotly*	Architected an interactive MVC data dashboard with custom Python CRUD modules, Dash/Plotly visualizations, and real-time geolocation mapping.
Machine Learning Models	*Python, TensorFlow, scikit-learn, NumPy, Matplotlib*	Trained and tuned a CNN for image classification and a Deep Q-Network (DQN) for reinforcement learning pathfinding — both implemented from scratch.
Super Mario Bros. 3D Engine	*C++, OpenGL, GLSL, GLM*	Recreated iconic 3D game scenes using custom GLSL shaders, manual mesh generation, and an optimized Phong lighting model rendering pipeline.
Product Design & UX	*Figma, Wireframes, User Research, Prototyping*	End-to-end UX design for mobile and web applications: user personas, lo-fidelity wireframes, hi-fidelity interactive prototypes, and design system documentation.

Full details, source code, and live demos at ***matthew-pool.github.io***

PROFESSIONAL EXPERIENCE

Lead Software Engineer & Product Designer

Jan 2025 – Present

AppBuddy (Independent Venture)

- **Product Strategy & Execution:** Architected and launched Flick, a production-ready native Android video streaming platform — designed end-to-end in Figma, built in Kotlin with clean MVVM architecture, and deployed to the Google Play Store.
- **Android Architecture:** Engineered a scalable native Android application using Jetpack Compose (Material Design 3), Hilt dependency injection, and offline-first data synchronization via DataStore.
- **ML-Driven Personalization:** Designed and deployed a proprietary recommendation engine using vectorization and cosine similarity, implementing an "exploration vs. exploitation" algorithm to dynamically rank video feeds while preventing echo chambers.
- **Infrastructure Optimization:** Integrated Retrofit with the Archive.org API, leveraging public domain assets to reduce content delivery costs to near-zero.
- **DevOps & Quality Assurance:** Built a robust CI/CD pipeline via GitHub Actions automating unit (JUnit/MockK) and UI tests (Espresso) to ensure virtually crash-free sessions in production.

Outbound Sales & Account Management (2004–2009)

Recurring Top Performer

Sales Representative • T.C.I.M. Services (B2B/B2C Outbound Sales)

Texas State Insurance License obtained independently within the branch.

- Conducted high-volume outbound sales calls on behalf of Fortune 500 clients including American Express, AT&T, and Citibank.
- Consistently ranked among the top performers in a rolling pool of 100+ representatives, earning "Top Performer of the Month" across multiple campaigns.
- **Training & Process Improvement:** Developed call scripts adopted across the team; trained junior sales representatives on objection handling and closing techniques.
- **Retention & Account Management:** Exceeded KPIs across acquisition, upsell, and retention metrics through effective communication and disciplined objection handling.

Technical Support, IT Operations & Inbound Sales (2011–2013)

Technical Support Specialist & Desktop Support • Sutherland / DirecTV / Frontier Communications

- **Technical Communication:** Translated complex hardware, software, VoIP, and ISP issues into clear guidance for diverse audiences.
- **Remote Diagnostics & Documentation:** Diagnosed and resolved complex issues via remote desktop tools; documented resolutions in CRM knowledge bases to improve team-wide efficiency.
- **Inbound Sales:** Identified upsell and upgrade opportunities during support interactions, consistently meeting or exceeding sales and satisfaction targets.

Shift Management & Customer Service (1997–2003 / 2009–2010) Management & Ops

Shift Manager & Retail Associate • Kmart / Fast Food & Customer Service (East TX)

- **Team Leadership & Operations:** Directed workflow, delegated tasks, and trained new hires across fast-paced retail and food service environments.
- **Autonomy & Trust:** Regularly trusted to close stores independently at night, maintaining full financial accountability for cash registers and daily deposits.
- Acted as the direct liaison between floor staff and upper management, successfully resolving escalated customer issues.

Independent Projects & Family Leave (2014–2021)

Construction & Primary Caregiver • Winnsboro, TX

- **Construction & Engineering:** Designed and constructed a residential structure (2-room tiny home) from raw materials — independently managed resource allocation, structural planning, and electrical/finish work.
- **Family Management:** Served as primary caregiver for a family of four while facilitating spouse's successful return to college, demonstrating planning, discipline, and long-term commitment.